

MINA SULTAN QABOOS, Oman

WMO: 412580

Lat: 23.628N

Lon: 58.572E

Elev: 10

StdP: 14.69

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 66.4	(c) 67.9	(d) 31.0	(e) 25.2	(f) 80.1	(g) 38.4	(h) 34.2	(i) 80.8	(j) 18.7	(k) 73.2	(l) 16.4	(m) 72.3	(n) 6.9	(o) 260	(p) 0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 7.2	(c) 104.6	(d) 72.2	(e) 102.6	(f) 72.3	(g) 100.6	(h) 72.7	(i) 88.3	(j) 91.0	(k) 87.1	(l) 90.2	(m) 85.9	(n) 89.5	(o) 8.1	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
87.7	201.8	90.2	86.4	192.7	89.1	85.1	184.6	88.5	54.1	91.1	52.3	90.2	50.8	89.4	94.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.5	(b) 14.3	(c) 12.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	WB	52.2	88.1	2.1	2.8	50.7	90.1	49.5	91.8	48.3	93.3	46.8	95.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	72.0	73.8	78.4	85.6	92.7	93.1	89.4	87.1	86.6	87.0	80.7	75.0
	DBStd	7.90	2.48	3.08	3.69	4.41	4.37	4.77	4.70	4.39	3.33	3.01	2.55	2.74
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12225	683	665	879	1067	1323	1294	1220	1151	1099	1148	920	776
	CDD65	6751	218	246	414	617	858	844	755	686	649	683	470	311
	CDH74	85438	321	789	3343	8275	13775	13553	11182	9529	8896	9548	4808	1419
(13)	CDH80	46047	1	94	689	4104	9312	9240	6719	5088	4587	5088	1084	41
(14)	Wind	WSAvg	6.2	6.1	6.4	6.4	6.2	6.3	6.4	6.8	7.1	6.1	5.5	6.0
(15) Precipitation	PrecAvg	3.5	1.0	1.1	0.6	0.2	0.0	0.0	0.0	0.2	0.0	0.0	0.2	0.4
	PrecMax	10.7	4.4	4.1	2.9	1.4	0.2	0.4	0.6	3.4	0.0	0.0	1.5	1.7
	PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.6	1.3	1.3	0.8	0.4	0.0	0.1	0.1	0.7	0.0	0.0	0.4	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.8	88.1	91.5	99.3	106.5	107.1	105.8	104.2	98.9	98.3	88.2	82.8
		MCWB	64.9	62.4	65.3	69.2	71.0	72.8	74.1	74.5	70.6	69.5	67.9	67.3
	2%	DB	77.8	82.2	88.4	96.7	103.7	104.9	102.7	99.8	95.8	95.8	86.5	81.0
		MCWB	66.6	67.1	65.6	67.2	70.0	72.1	74.8	75.6	73.2	70.0	70.9	68.1
	5%	DB	76.8	79.7	85.5	94.3	101.7	103.0	99.2	95.5	93.1	93.6	85.3	79.9
		MCWB	67.2	66.9	66.9	67.4	70.5	72.5	76.7	77.8	74.9	70.8	71.9	68.7
	10%	DB	75.8	77.8	83.5	92.1	99.8	100.9	95.6	92.5	91.1	91.7	84.3	78.8
		MCWB	66.7	67.6	67.4	67.8	70.8	73.8	79.9	79.2	77.6	72.5	72.2	69.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.9	74.9	78.8	82.0	87.5	90.2	89.4	87.9	86.4	85.1	79.1	76.2
		MCDB	76.0	76.7	80.4	85.2	89.8	92.1	92.1	89.3	88.7	87.7	83.5	78.8
	2%	WB	71.5	73.9	76.8	79.9	85.9	89.0	87.6	86.2	84.5	83.4	77.3	74.6
		MCDB	74.9	76.3	79.9	84.1	89.7	91.5	90.4	88.6	87.4	86.9	83.1	78.4
	5%	WB	69.9	72.6	75.1	78.6	84.4	87.8	86.4	84.9	83.4	81.9	76.0	73.1
		MCDB	75.1	76.1	79.0	84.0	89.7	91.1	89.2	87.6	86.8	86.8	82.9	78.0
	10%	WB	68.7	70.9	73.5	77.3	82.8	86.3	85.5	83.7	82.5	80.6	74.8	71.1
		MCDB	74.8	75.9	79.1	84.0	89.6	91.0	88.9	87.1	86.4	86.8	82.5	77.4
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.0	5.6	6.8	8.1	7.2	5.6	5.6	5.2	5.5	4.6	4.7
		MCDBR	5.6	7.2	8.9	9.7	11.5	11.3	11.0	11.5	9.3	9.4	5.3	5.5
		MCWBR	5.3	7.5	9.0	9.0	9.9	11.8	10.0	9.0	9.9	11.0	6.9	6.0
	5% WB	MCDBR	4.8	4.6	6.1	6.8	7.4	7.0	6.7	5.8	6.0	4.6	4.6	4.7
		MCWBR	4.9	6.4	8.5	9.2	10.3	10.3	6.8	6.7	6.7	8.5	6.3	5.5
(40) Clear-Sky Solar Irradiance	taub		0.422	0.465	0.525	0.572	0.596	0.669	0.740	0.678	0.577	0.478	0.440	0.422
	taud		2.242	2.089	1.918	1.837	1.788	1.674	1.587	1.686	1.852	2.063	2.198	2.231
	Ebn at Noon		263	259	249	240	233	215	200	213	233	252	255	258
	Edh at Noon		40	49	61	68	71	79	86	78	65	50	41	39
(44) All-Sky Solar Radiation	RadAvg		1402	1659	1931	2152	2354	2293	2241	2086	1970	1779	1478	1328
(45)	RadStd		66	72	97	107	49	78	92	58	39	45	44	67

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.41	N/A	N/A	N/A	N/A	+0.70	+0.81	N/A	N/A	+173